

Phylogenomics

Erkison Odih

DEGAiN Bioinformatics Workshop

23 July 2026

Good stuff ?

- Cilia Antoniou
- Adam Leache
- Zoe Dyson
- Other online sources

Bad stuff?

Mine!



Phylogenomics

The inference of evolutionary relationships between different biological entities using genomic data.



Taxonomy



Forensics and outbreak
investigations

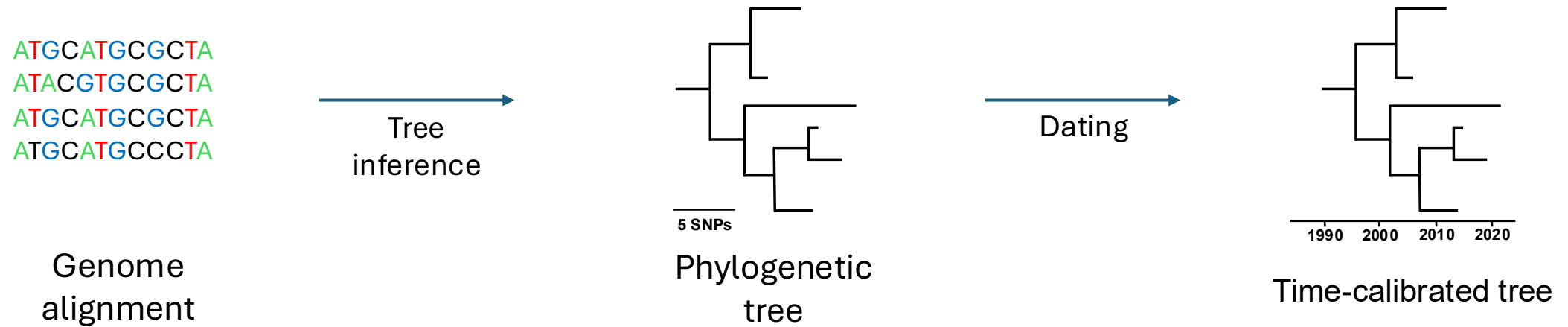


Pathogen and
antimicrobial
resistance surveillance



Drug development

Phylogenomics workflow



- **Molecular dating:** translating genetic distance between taxa into chronological distance

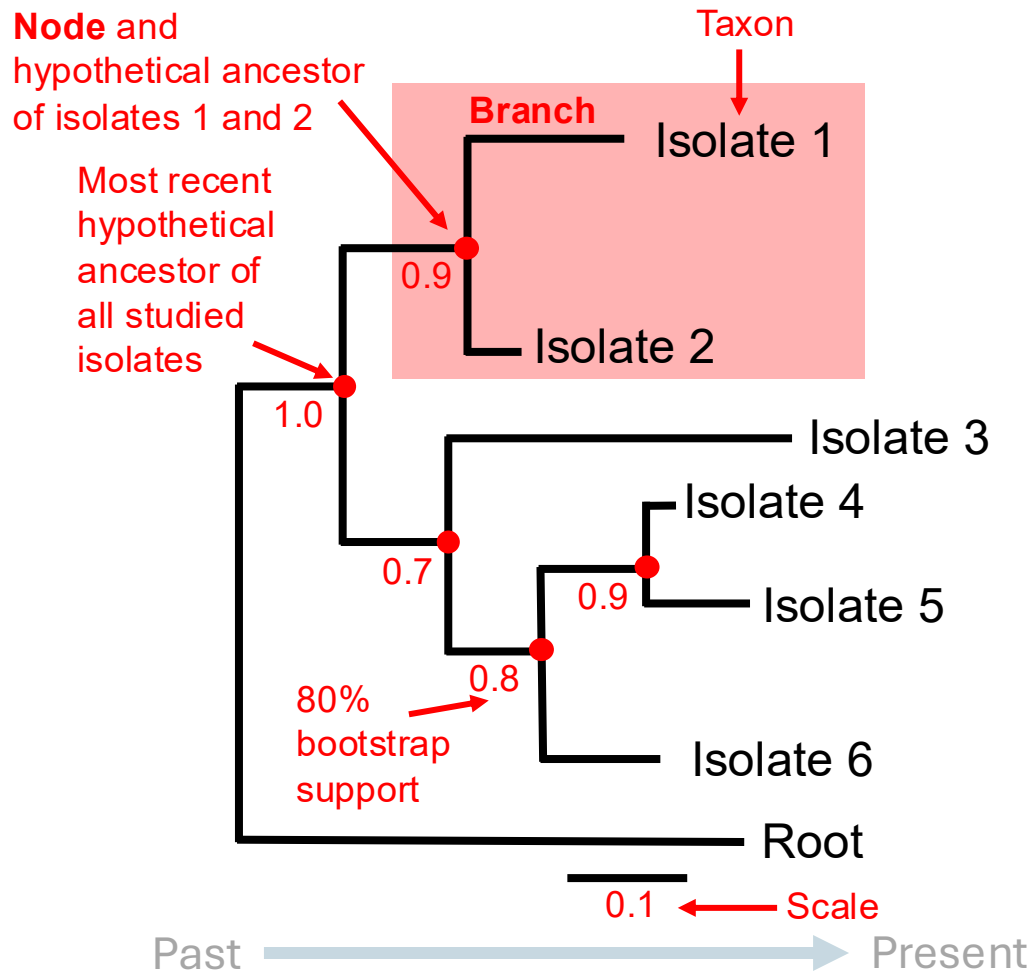
Phylogenetic trees

- Graphical representations of the relationships between taxa
- Can be estimated from any heritable, potentially variable, observable feature of an organism

Information encoded in a phylogenetic tree

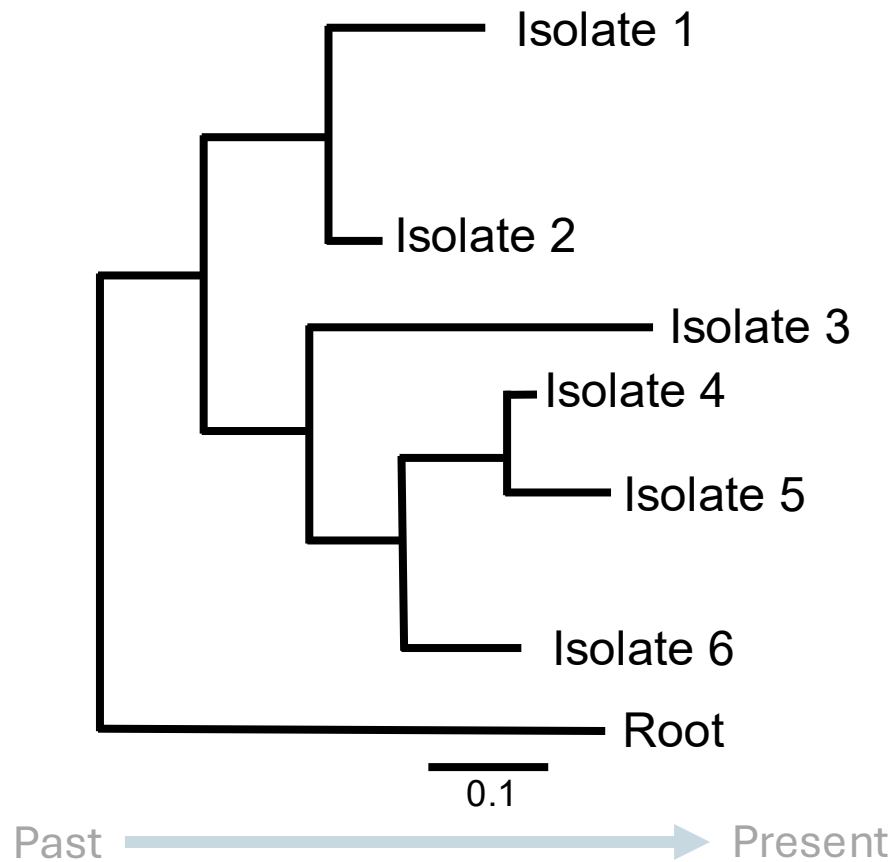
- Topological information: *who is most closely related to whom?*
- Evolutionary divergence: *how much change has occurred along branches?*
- Evolutionary history: *what happened along the branches of the tree?*

Parts of a phylogenetic tree



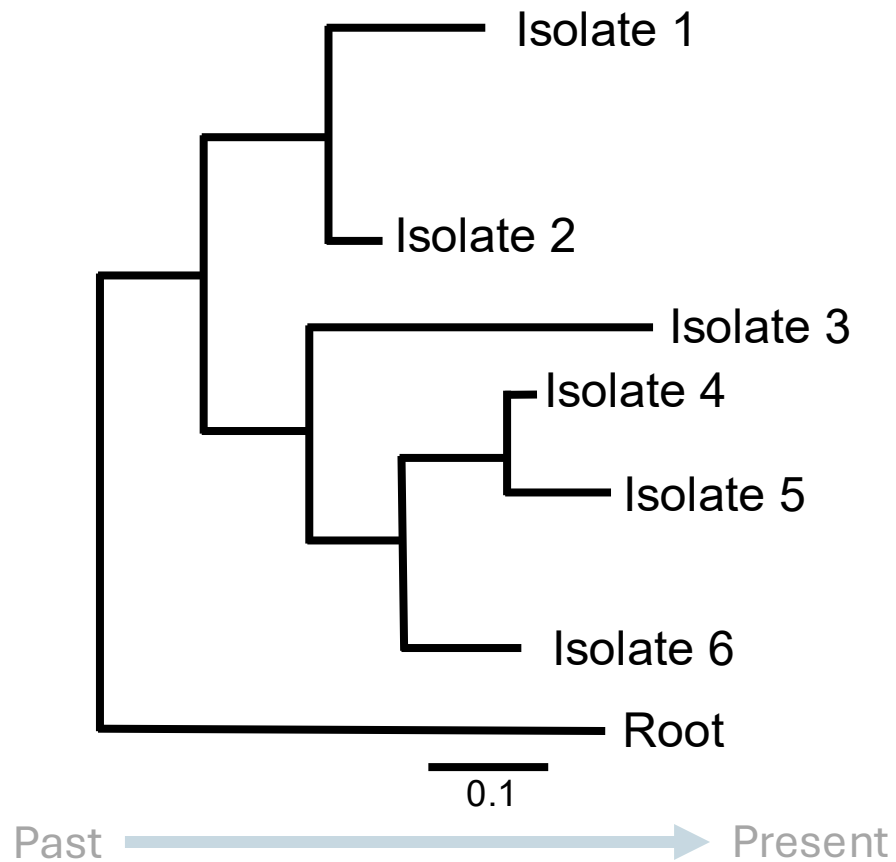
- **Taxa** are on the tree tips/leaves
- **Nodes:** hypothetical taxa ancestors
- **Branches** connect nodes and taxa
- **Topology:** branching structure of a tree
- **Clade:** taxa sharing a unique common ancestor
- **Bootstraps:** proportion of multiple replicate trees supporting each node.
- **Root:** most distantly related to all taxa on the tree; tells direction of evolution

Quick exercise 1



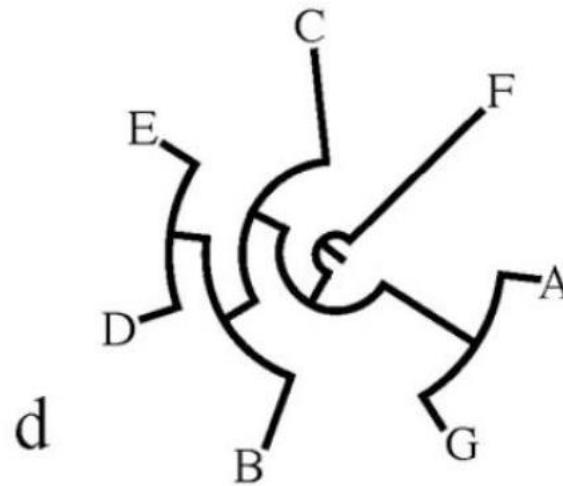
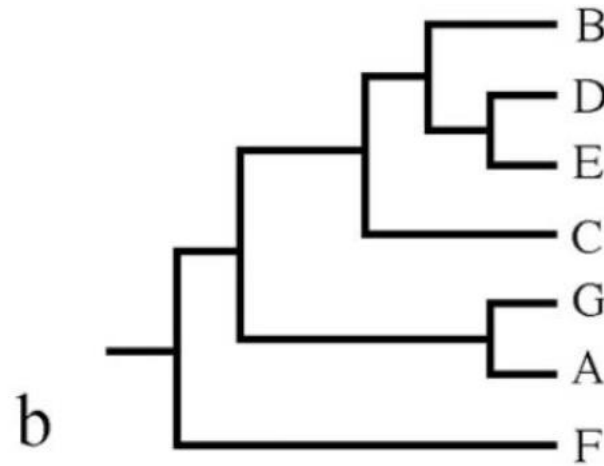
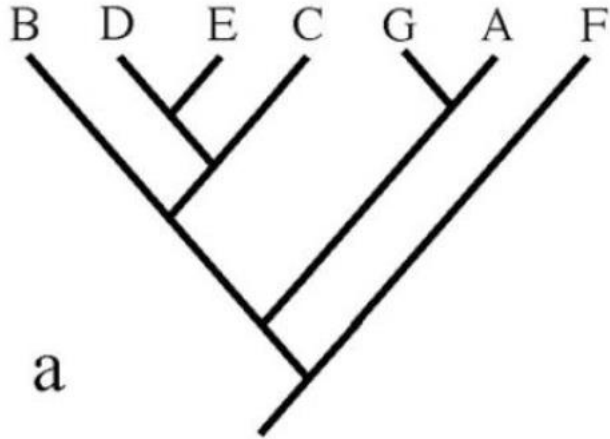
Is Isolate 4 more closely related to Isolate 3 or Isolate 5?

Quick exercise 2



Excluding the root, which isolate is most distantly related to Isolate 2?

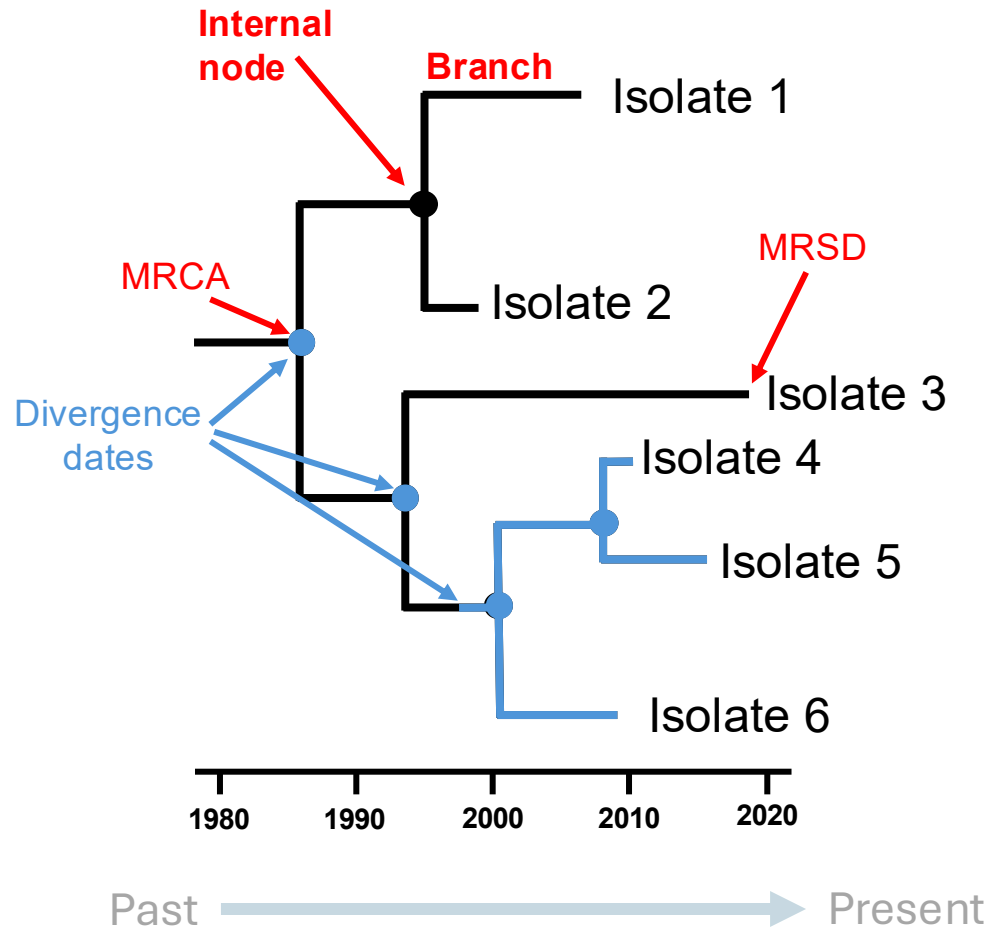
Quick exercise 3



Which of these four trees depict the same evolutionary history?

(ignore branch lengths)

Dated phylogenetic trees



- **MRCA:** most recent common ancestor of all nodes
- **tMRCA:** time since any two or more individuals shared a common genetic ancestor
- **Divergence date:** also emergence or existence date. The date a lineage or specific trait emerged/diverged/existed. Could also be along any branch
- **MRSD:** most recent sampling date

Molecular clock models

<https://beast.community/clocks>

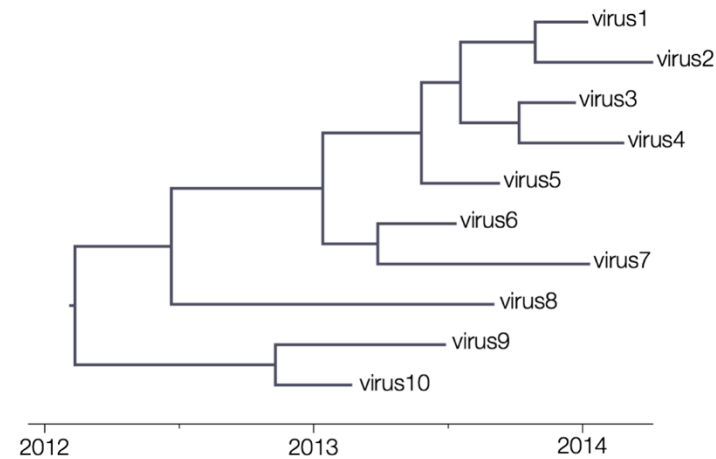
Drummond et al (2006). *PLoS Biology*

Lepage et al. (2007). *Molecular Biology and Evolution*

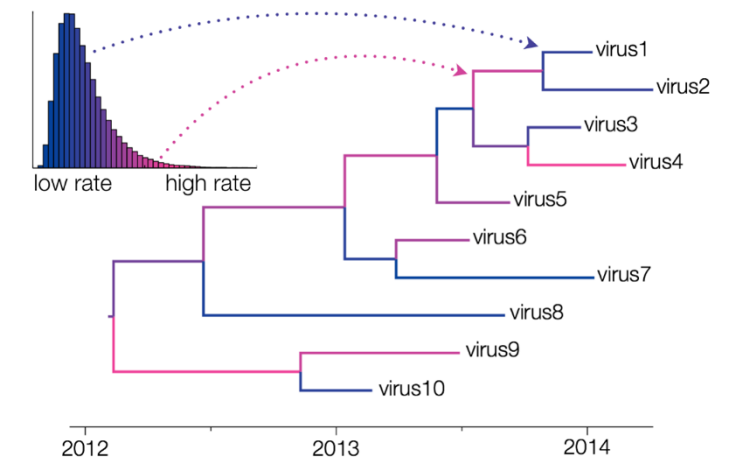
- Molecular clock models underpin molecular evolution analyses; it relies on the assumption that genetic change accumulates at a consistent rate proportional to calendar time
- Molecular dating relies on this **temporal signal**, i.e. sufficient genetic change in the dataset over time to allow the measurement of this genetic change-time relationship
- During dating, clocks are calibrated using sampling dates and genomic change (distance) data

Common models

- **Strict clock:** this model assumes that evolution occurs at the same rate throughout the tree
- **Relaxed clock:** this model assumes that evolutionary rates vary across the branches of the tree



Under the strict clock model, evolution occurs at the same rate at every branch.



Under the uncorrelated relaxed clock model, the evolutionary rate at each branch is drawn from an underlying distribution.

Bayesian inference

- Molecular dating is commonly done using **Bayesian inference** (based on Bayes' Theorem)
- This is an **iterative** statistical method that uses probability to quantify and update beliefs about certain **parameters** and their distribution as new **evidence** emerges
- Bayesian probability is unlike long-run probability in the frequentist approach

- | | |
|--|---------------|
| • “If I toss this coin 100 times, 50% will be heads” | – Frequentist |
| • “It’s 50% likely that the <u>next</u> coin-toss will be heads” | – Bayesian |

First, a crash course on probability

A record of sports preferences among 100 people

	Male	Female	TOTAL
Football	16	24	40
Tennis	20	5	25
Other	10	25	35
TOTAL	46	54	100

	Male	Female	TOTAL
Football	0.16	0.24	0.40
Tennis	0.20	0.05	0.25
Other	0.10	0.25	0.35
TOTAL	0.46	0.54	1

Marginal probability
distribution

**adds up to 1*

Marginal probability

$$P(\text{Football}) = ?$$

$$P(\text{Tennis}) = ?$$

First, a crash course on probability

A record of sports preferences among 100 people

	Male	Female	TOTAL
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Other	0.10	0.25	0.35
TOTAL	0.46	0.54	1

Marginal probability
distribution

**adds up to 1*

Marginal probability

$$P(\text{Football}) = 0.4$$

$$P(\text{Tennis}) = 0.25$$

First, a crash course on probability

A record of sports preferences among 100 people

	Male	Female	TOTAL
Football	16	24	40
Tennis	20	5	25
Other	10	25	35
TOTAL	46	54	100

	Male	Female	TOTAL
Football	0.16	0.24	0.40
Tennis	0.20	0.05	0.25
Other	0.10	0.25	0.35
TOTAL	0.46	0.54	1

Joint probability
distribution

**adds up to 1*

Joint probability

Independent: $P(A \cap B) = P(A) \times P(B)$

Dependent: $P(A \cap B) = P(A) \times P(B|A)$

$P(\text{Male} \cap \text{Football}) = ?$

First, a crash course on probability

A record of sports preferences among 100 people

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Football	16	24	40
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Joint probability
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**adds up to 1*

Joint probability

Independent: $P(A \cap B) = P(A) \times P(B)$

Dependent: $P(A \cap B) = P(A) \times P(B|A)$

$$P(\text{Male} \cap \text{Football}) = 0.16$$

First, a crash course on probability

A record of sports preferences among 100 people

	Male	Female	TOTAL
Football	16	24	40
Tennis	20	5	25
Other	10	25	35
TOTAL	46	54	100

	Male	Female	TOTAL	P(sport F)
Football	0.16	0.24	0.40	0.444
Tennis	0.20	0.05	0.25	0.093
Other	0.10	0.25	0.35	0.463
TOTAL	0.46	0.54	1	

Conditional probability
distribution

**adds up to 1*

Conditional probability $P(A|B) = \frac{P(A \cap B)}{P(B)}$

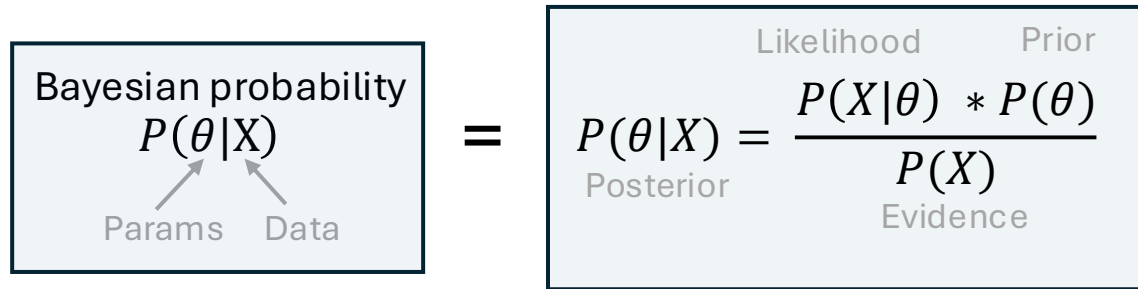
$$P(\text{Football}|\text{Female}) = \frac{0.24}{0.54} = 0.444$$

$$P(\text{Other}|\text{Female}) = \frac{0.25}{0.54} = 0.463$$

$$P(\text{Tennis}|\text{Female}) = \frac{0.05}{0.54} = 0.093$$

Bayesian inference

- Bayesian inference builds on conditional probability



The diagram illustrates the Bayesian inference formula. On the left, a box labeled 'Bayesian probability' contains the expression $P(\theta|X)$. Below this expression, 'Params' has an arrow pointing up to θ , and 'Data' has an arrow pointing up to X . This box is followed by an equals sign and another box. The second box contains the formula $P(\theta|X) = \frac{P(X|\theta) * P(\theta)}{P(X)}$. Above $P(X|\theta)$ is the label 'Likelihood', and above $P(\theta)$ is the label 'Prior'. Below the denominator $P(X)$ is the label 'Evidence'. Below the entire fraction is the label 'Posterior'.

$$\text{Bayesian probability } P(\theta|X) = \frac{\text{Likelihood } P(X|\theta) * \text{Prior } P(\theta)}{\text{Evidence } P(X)}$$

Posterior

- Prior probability (prior):** probability distribution based on initial belief or existing knowledge (assigned to each model parameter)
 - Likelihood:** The probability of observing the newly collected data, assuming initial hypothesis is true.
 - Evidence (marginal likelihood):** normalizing constant representing the overall probability of observing the data across (or irrespective of) all possible hypotheses.
 - Posterior probability (posterior):** The updated probability based on new data. This becomes the new prior for any future data collection.
- Bayesian inference involves an iterative process of taking prior beliefs (prior) of the probability distribution of a parameter and then updating that belief (posterior) based on observed data.
 - In real-world applications, parameter complexity is such that it is impossible to calculate the marginal likelihood

Markov Chain Monte Carlo (MCMC)

- MCMC helps overcome the intractable denominator problem by approximating the true posterior distribution
- MCMC iteratively draws samples from the unnormalised posterior distribution, comparing the previous sample to a proposed candidate sample (accepted or rejected).
- The goal is for the chain to converge to a stationary distribution that is identical to the target posterior distribution
- Separate concepts to consider:
 - ☐ Monte Carlo
 - ☐ Markov Chain
 - ☐ Guided random walks (Metropolis-Hastings algorithm)
 - ☐ Burn-in
 - ☐ Autocorrelation and thinning

Markov Chain Monte Carlo (MCMC)

Monte Carlo

- Statistical method used to model the probability of different outcomes in complex systems
- Follows the “law of large numbers” using repeated independent random sampling
- By averaging a large number of random samples, the sample mean converges to the true population mean with high precision.

```
# approximate the distribution of student heights
n <- 10
x <- rnorm(n, mean = 175, sd = 6)
hist(x, breaks = 50, probability = TRUE,
      col = "lightblue", border = "white")
# repeat with n = 100000
```

Markov Chain

- A Markov Chain is a **stochastic** process describing a **sequence of events** where the transition to the next state depends strictly on the current state, independent of the past path.

Markov Chain Monte Carlo (MCMC)

Metropolis-Hastings Algorithm

Proposal

Generate a candidate sample based on the proposal distribution.

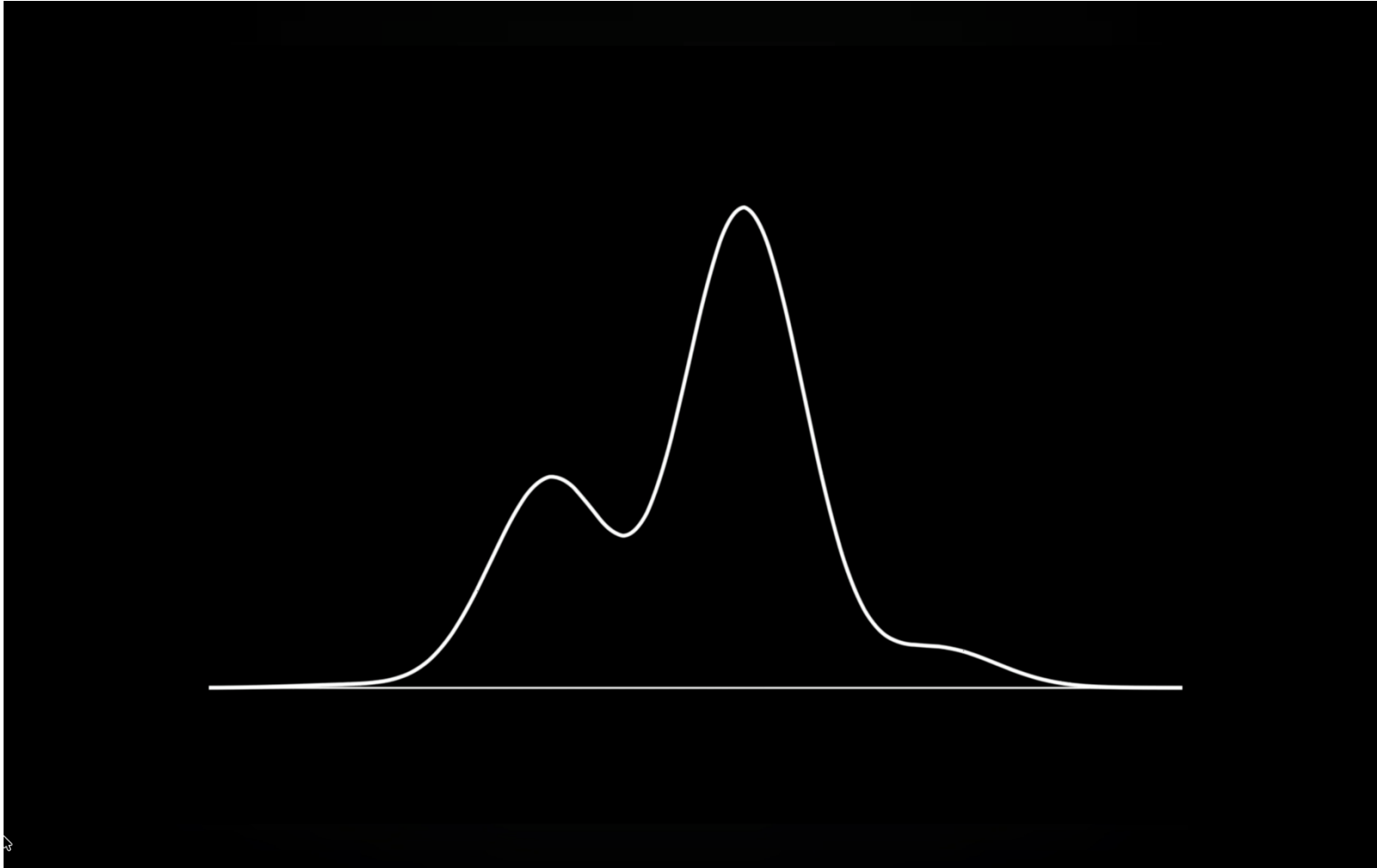
Acceptance/Rejection

Compare the relative densities; if the move is "uphill," always accept. If "downhill," accept probabilistically.

Convergence

Repeat until the sequence of samples converge around a stationary distribution.

Markov Chain Monte Carlo (MCMC)



<https://www.youtube.com/watch?v=3qodjHRUxAo>

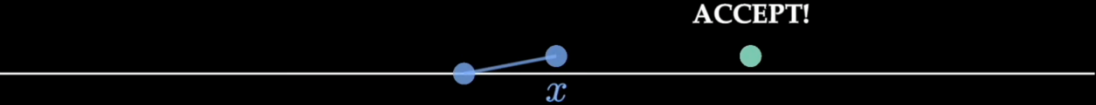
Markov Chain Monte Carlo (MCMC)

Markov Chain Monte Carlo Explained in 10 Minutes

Chain Creation Rules

If $f(x') > f(x)$:
Accept x'

If $f(x') < f(x)$:
Accept with prob. $\frac{f(x')}{f(x)}$



ACCEPT!

x

4:08 / 10:44

<https://www.youtube.com/watch?v=3qodjHRUxAo>

Markov Chain Monte Carlo (MCMC)

Interactive MCMC visualisation: <https://chi-feng.github.io/mcmc-demo/app.html>

BactDating

Bayesian inference of ancestral dates on bacterial phylogenetic trees

[Xavier Didelot](#) , [Nicholas J Croucher](#), [Stephen D Bentley](#), [Simon R Harris](#), [Daniel J Wilson](#)

Nucleic Acids Research, Volume 46, Issue 22, 14 December 2018, Page e134,
<https://doi.org/10.1093/nar/gky783>

GitHub:

<https://github.com/xavierdidelot/BactDating>

BactDating

Estimated parameters

- **Mu (μ):** mean substitution rate
- **Sigma (σ):** the standard deviation of the clock rate across branches
- **Alpha (α):** coalescent time unit
- **Node dates**

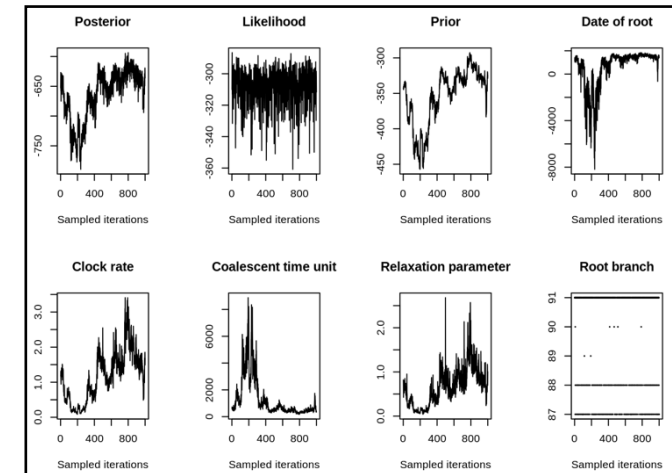
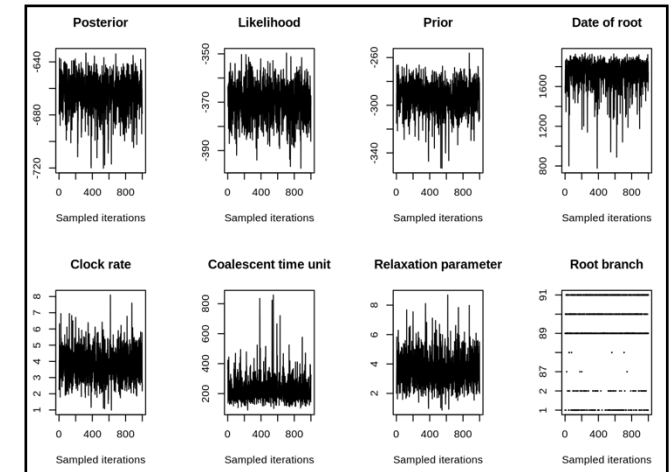
Target distribution:

$$p(\mathcal{T}, \Theta | \mathcal{P}) \propto p(\mathcal{P} | \mathcal{T}, \Theta) p(\Theta) p(\mathcal{T} | \Theta)$$

Assessing MCMC convergence

Trace plots

- These show the distribution of the parameter estimate(s) across sampled points in the chain
- Allows visual assessment of:
 - Stationarity and convergence: parameters fluctuate around a constant mean and variance
 - Mixing: the chain rapidly and frequently moves across the parameter space
 - Trending (upward or downward)
 - Non-overlapping chains: indicates when different chains are trapped in local minima or maxima
- “Fuzzy caterpillar” indicates a well-mixed and converged chain



Assessing MCMC convergence

Effective sample size (ESS)

- ESS refers to the number of effectively independent draws from the posterior distribution (i.e., a measure of autocorrelation)
- Low ESS values indicate poor mixing of the MCMC chain
- Recommended acceptable threshold: >100 for all parameters of interest
- ESS may be improved by:
 - Increasing chain length
 - Increasing sampling frequency (more thinning)

Gelman-Rubin diagnostic

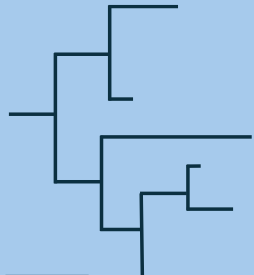
- This is a numerical convergence summary (potential scale reduction factor) of multiple chains
- Ratio of between-chain to within-chain variance
- Typically, 2 or more independent chains are run and compared
- Acceptable values: PSRF <1.2

Overview of Dating Analysis

1

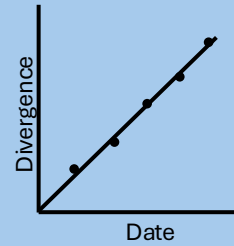
Tree
inference

5 SNPs

A phylogenetic tree diagram showing branching relationships between several taxa. A scale bar at the bottom left indicates a distance of 5 SNPs.

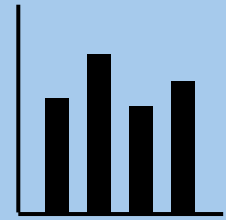
2

Root-to-tip
regression



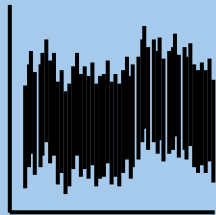
3

Model
selection



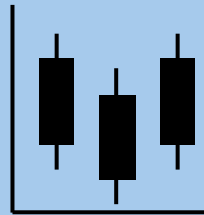
4

Convergence
assessment



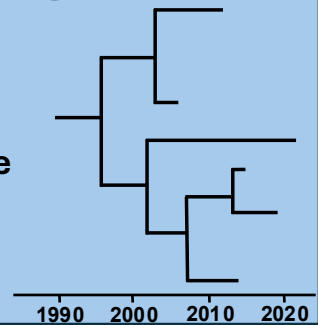
5

Test
temporal
signal



6

Dated tree



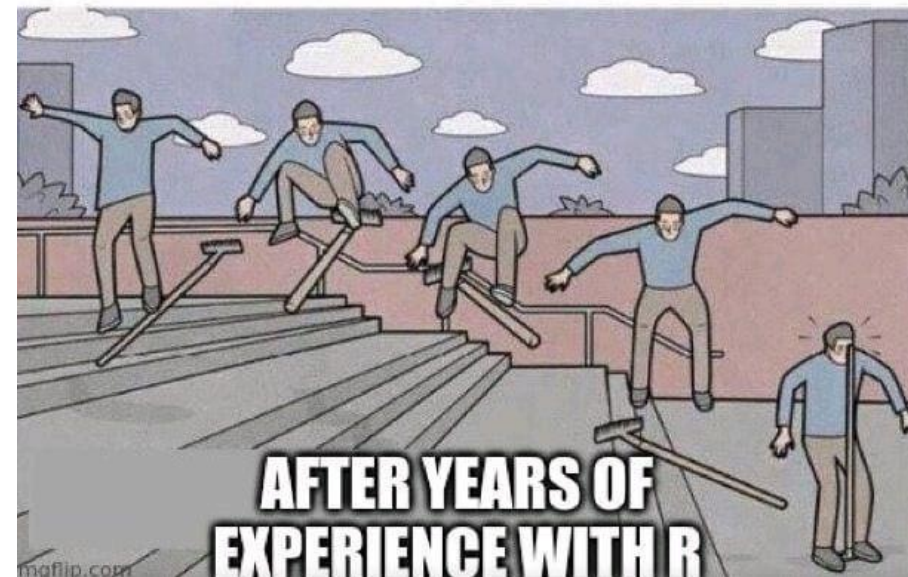
Now to some actual analyses

- Formatting trees and metadata for BactDating
- Assess temporal signal
- Perform BactDating MCMC
- Model comparison
- Diagnostics
- Visualising and interpreting the dated phylogeny

Group 1: Cluster 12

Group 2: Cluster 17

Group 3: Cluster 20



Practical session



Improving temporal signal in the data

Before running endless iterations, consider ways you may improve the temporal signal in the data:

- General dataset evaluation
 - Number of genomes
 - Date range and variance
 - Missing dates?
- Tree rooting
- Exclude divergent genomes
- Improve recombination detection and removal (alternate tools, alternate parameters)

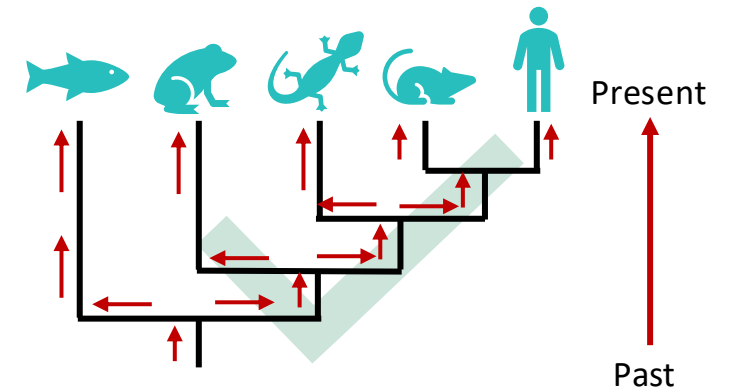
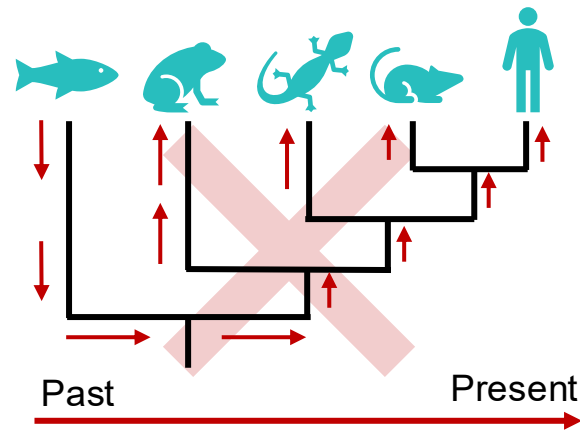
Vertical: A fast and robust tool for recombination detection in large-scale bacterial genomic datasets

 Erkison Ewomazino Odih, Ryan R Wick, Kathryn E Holt

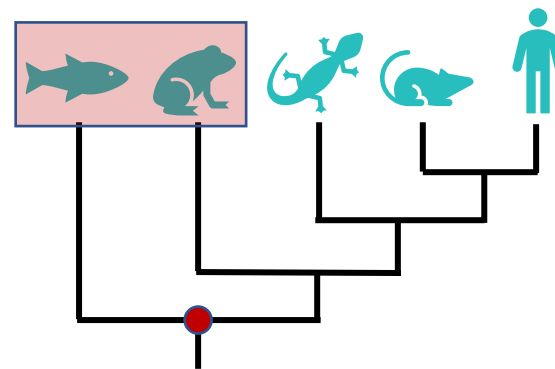
doi: <https://doi.org/10.64898/2026.04.21.719734>

Phylogenetic tree interpretation - Misconceptions

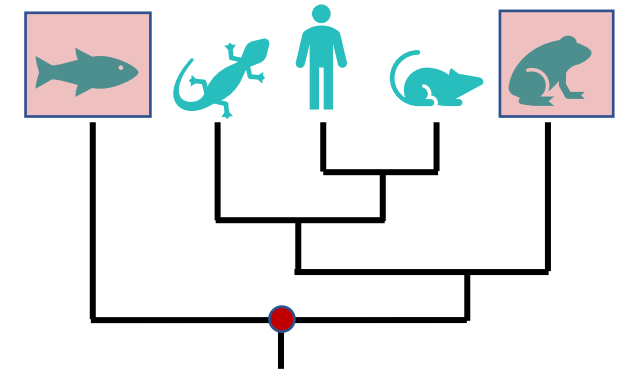
1. Evolutionary timeline flows from tips/leaves



2. Related species have proximal leaves



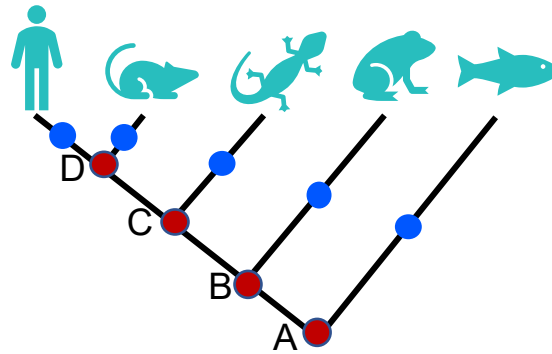
=



Relationships are inferred from common ancestry, not leaf proximity

Phylogenetic tree interpretation - Misconceptions

3. Genetic change occurs only at nodes



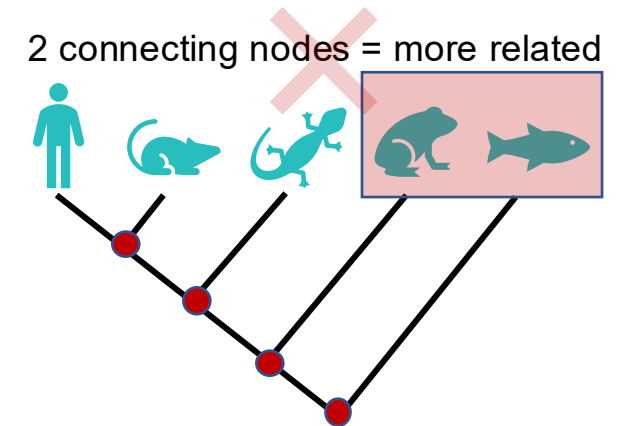
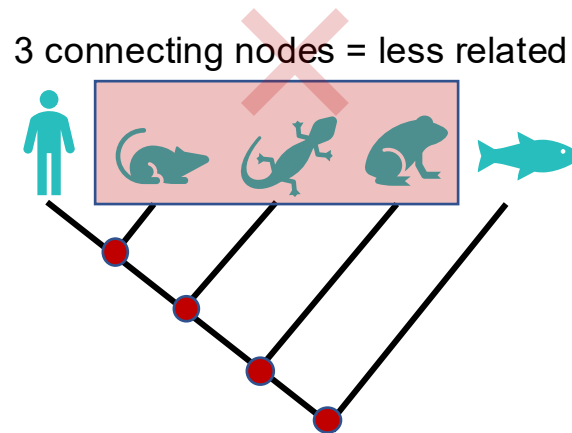
Fact:

- Changes accumulate in taxa over time and are present along branches
- Nodes simply represent (hypothetical) common ancestors between taxa

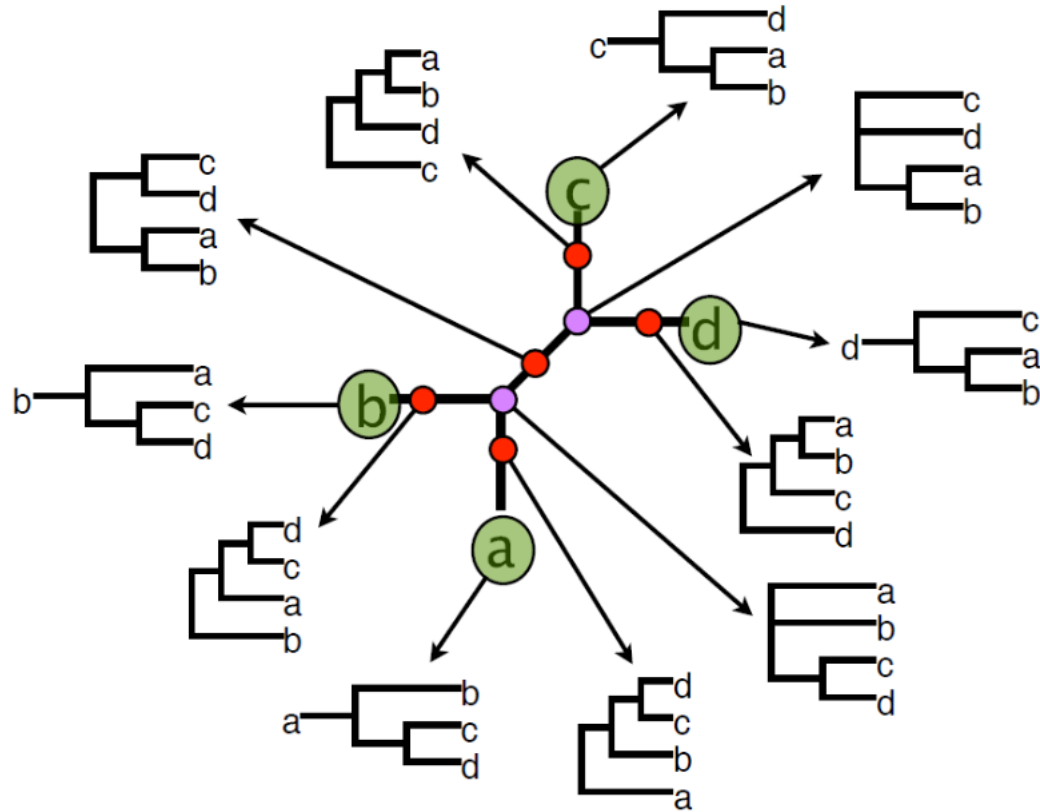
● - Shared ancestor; divergence event

● - Unique ancestor

4. Related species have fewer connecting nodes



Tree rooting



Unrooted trees can be rooted at the:

- Branches
- Internal nodes
- External nodes (tips)

Bacterial phylodynamics

Erkison Odih

DEGAiN Bioinformatics Workshop

24 July 2026

Phylodynamics

Phylodynamics is a blend of phylogenomics and epidemiology to study the evolution and spread of biological entities and their features of interest.



Population dynamics



Outbreaks and
transmission
dynamics

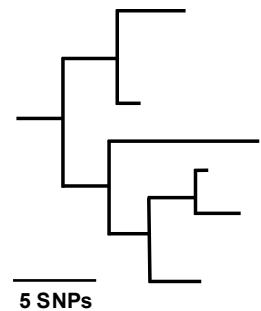


Pathogen emergence
and geographic
spread



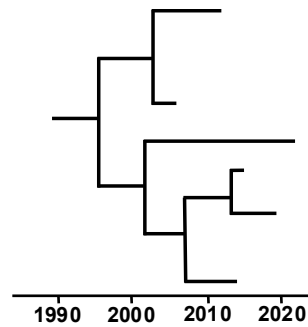
AMR evolution

Phylodynamics workflow



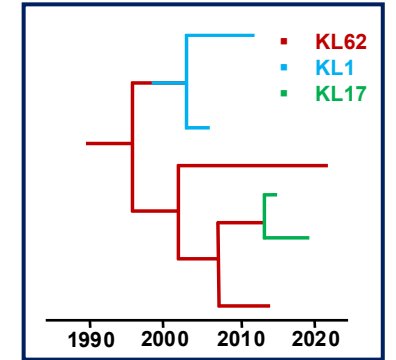
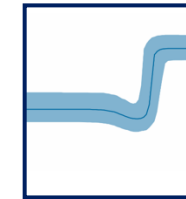
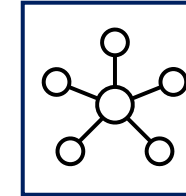
Phylogenetic tree

Dating



Time-calibrated tree

Modelling traits and params of interest



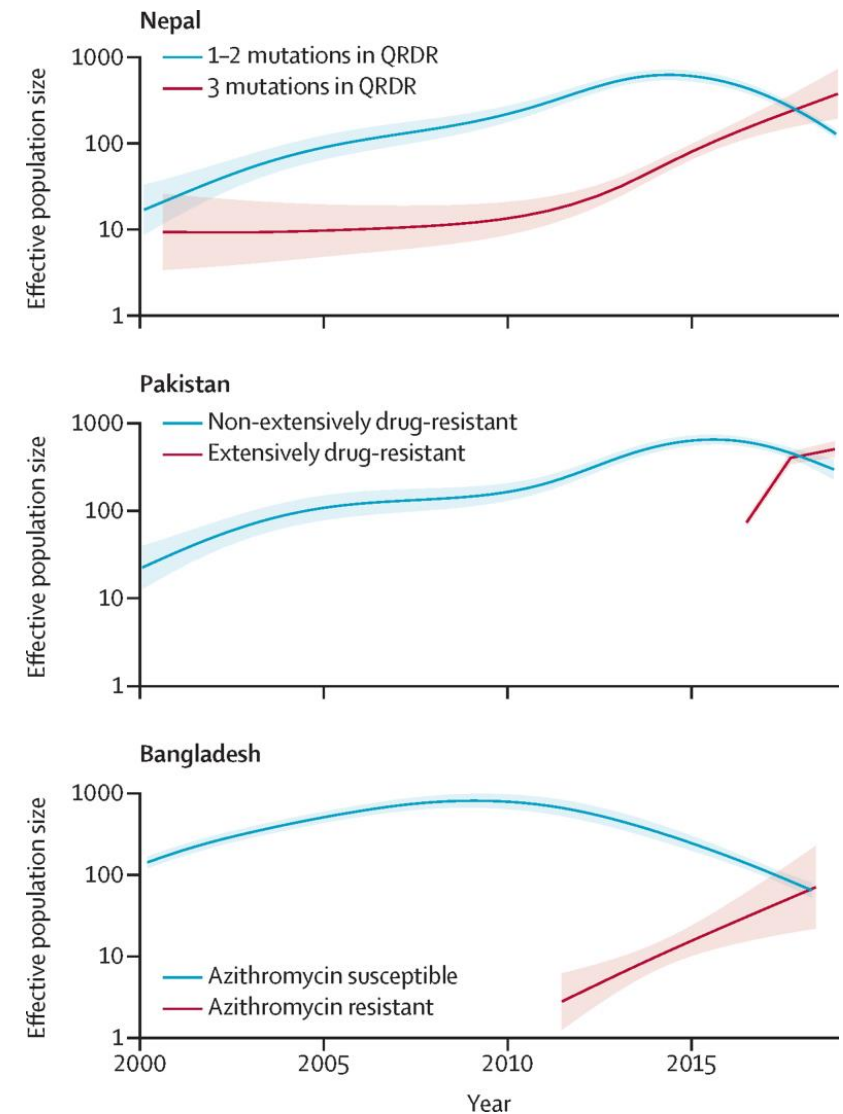
- Transmission modelling
- Infer population dynamics
- Ancestral state reconstruction

Phylodynamics applications

- Effective population size
- Basic reproduction number
- Transmission dynamics
- Ancestral state reconstruction

Effective population size (N_e)

- Effective population size is the size of an idealised population that would experience the same rate of genetic drift as the actual population.
- The rate of genetic drift is not proportional to census population size (N_c) but to the effective population size (N_e). As much genetic variation is captured in this effective population as the census population
- These analyses are thus sensitive to representative sampling
- In phylodynamics, we are interested in modelling the changes in effective population size over time, e.g. the expansion of certain lineages
- Skygrowth R package



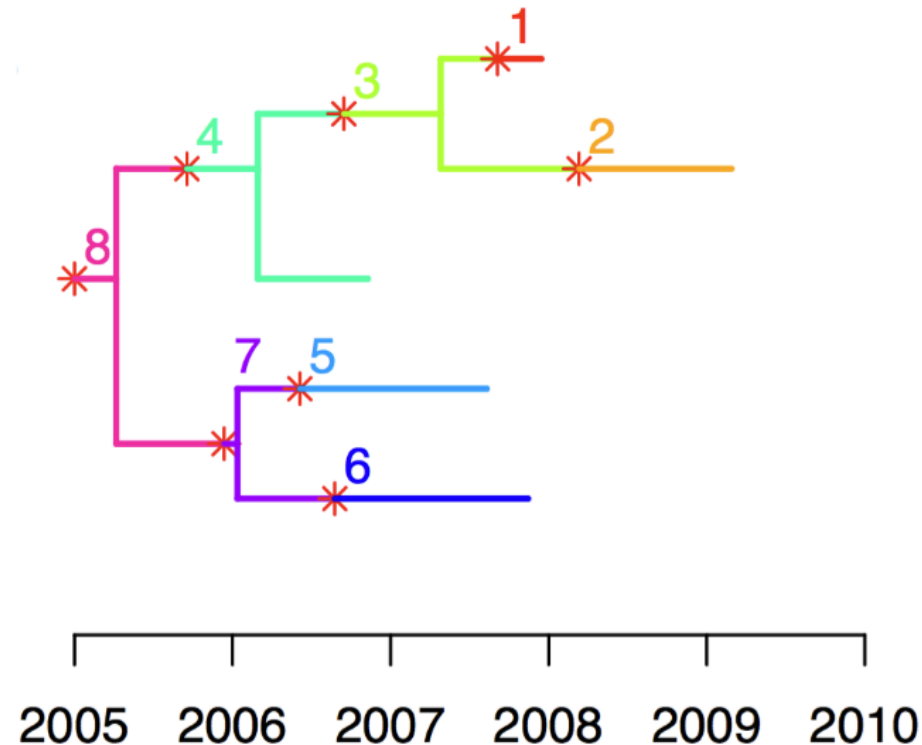
DaSilva et al (2022). *The Lancet Microbe*

Basic reproduction number (R_0)

- R_0 is a measure of a pathogen's contagiousness.
- R_0 is the expected number of cases directly generated by one case in a susceptible population
- R_0 is however influenced by environmental and other factors so needs to be interpreted carefully
- Can be modelled based on dated phylogenies to, for example, **compare R_0 across lineages** of interest or **model changes in R_0 over time**
- Skygrowth R package

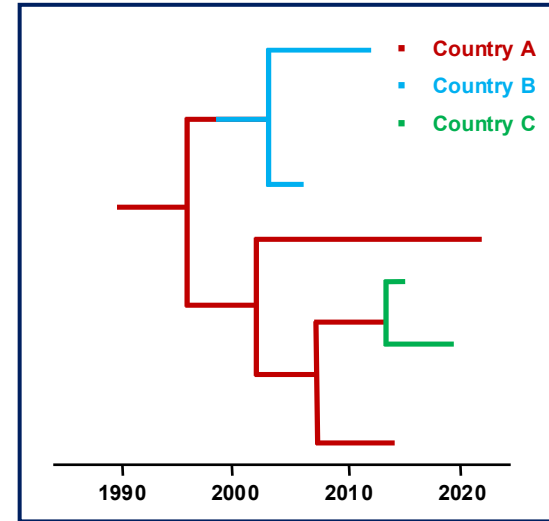
Transmission dynamics

- Transmission dynamics, including the direction and sources of infections, can be inferred from dated trees
- TransPhylo R package
- Accounts for within-host genetic diversity
- “The main output is a transmission tree which indicates who infected whom, including the potential existence of unsampled individuals who may have acted as missing transmission links”



Ancestral state reconstruction

- This is the reconstruction of discrete or continuous traits along the evolutionary history of the studied taxa
- When applied to a dated phylogenetic tree, the dates of specific emergence or acquisition events can be estimated.
- **Discrete** and **continuous** states can be mapped parsimoniously or using maximum likelihood or Bayesian approaches with different models (Mk model; Brownian motion model)



Ancestral state reconstruction – Discrete traits

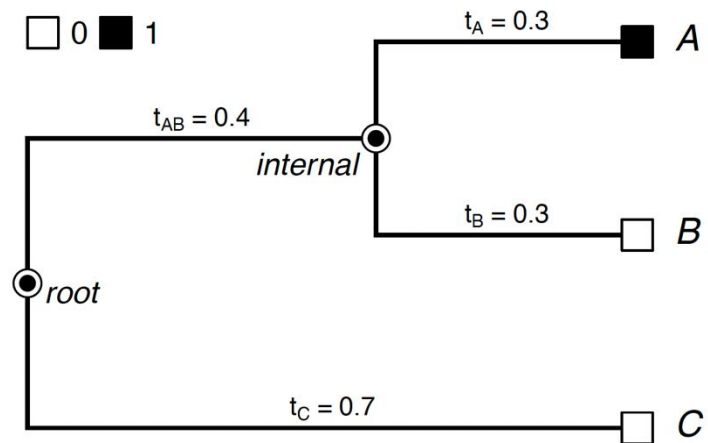
The ‘Mk’ model for discrete character ASR

- Continuous time Markov chain with k possible states
- Fully parametrized by a $k \times k$ matrix (Q) representing the **transition rates** between states

$$P_t = \exp(Qt),$$

where t = time (unit edge length), and each element of $P_t = P_{i,j}$ and gives $P(j|i, t)$

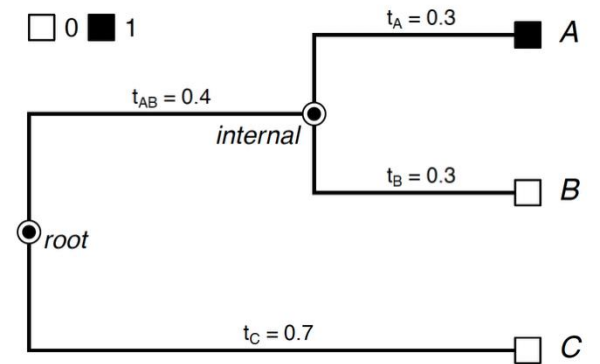
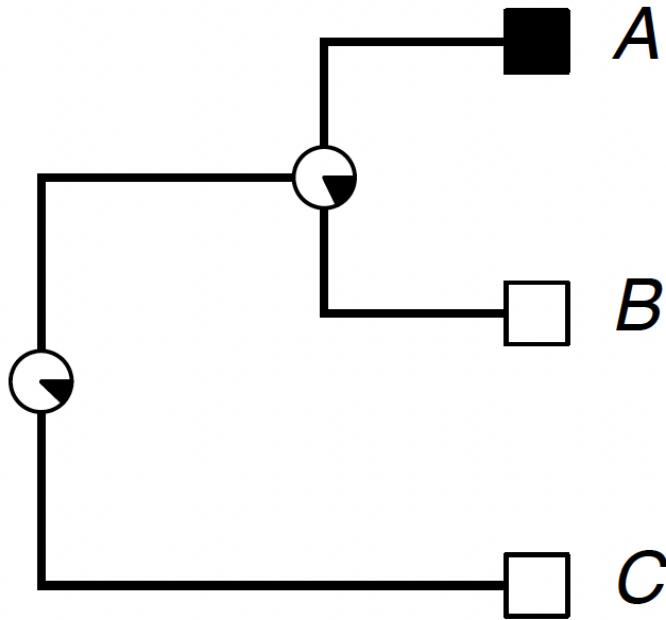
- Q is typically estimated as the value that maximises the likelihood of the data, given the tree.



Ancestral state reconstruction – Discrete traits

Marginal ancestral state estimation

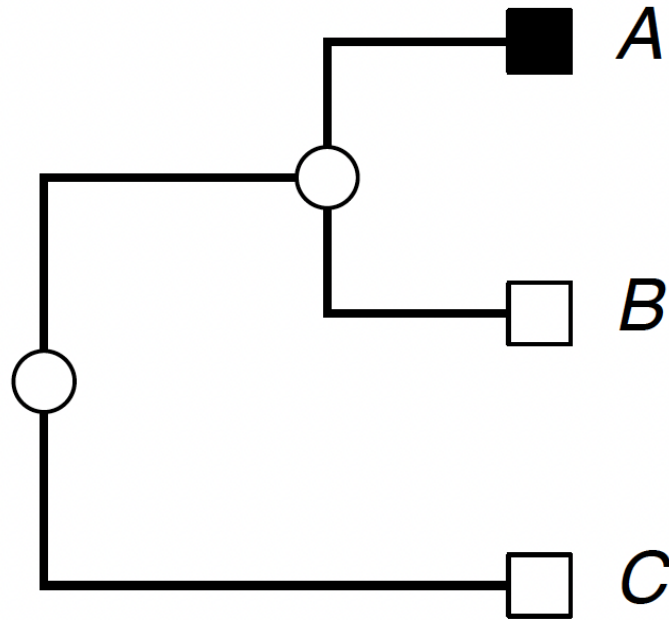
- Tree is traversed, and each internal node is visited
- For each visited node, we compute the probability of observing the data on the tips, given that the node is in each of the set of distinct trait values.



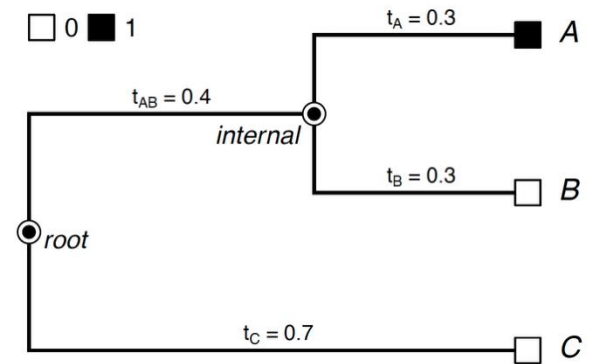
Ancestral state reconstruction – Discrete traits

Joint ancestral state estimation

- Here we find the combination/set of states at all internal nodes that jointly maximize the probability of our observed tip states



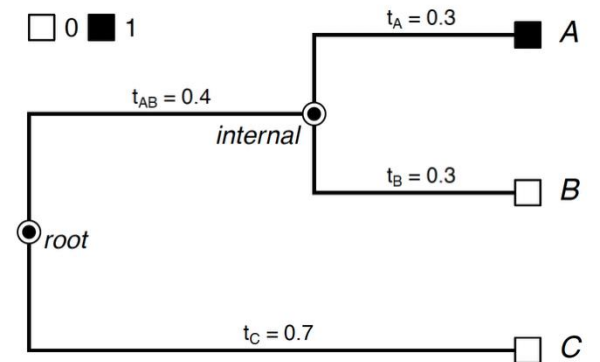
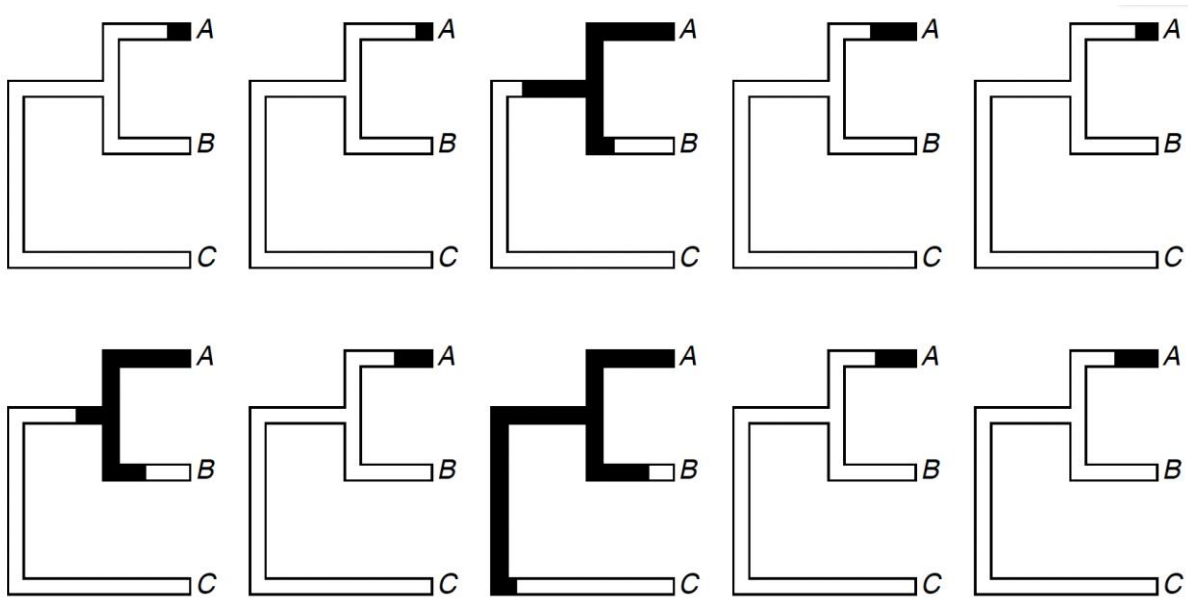
Revell and Harmon, 2022



Ancestral state reconstruction – Discrete traits

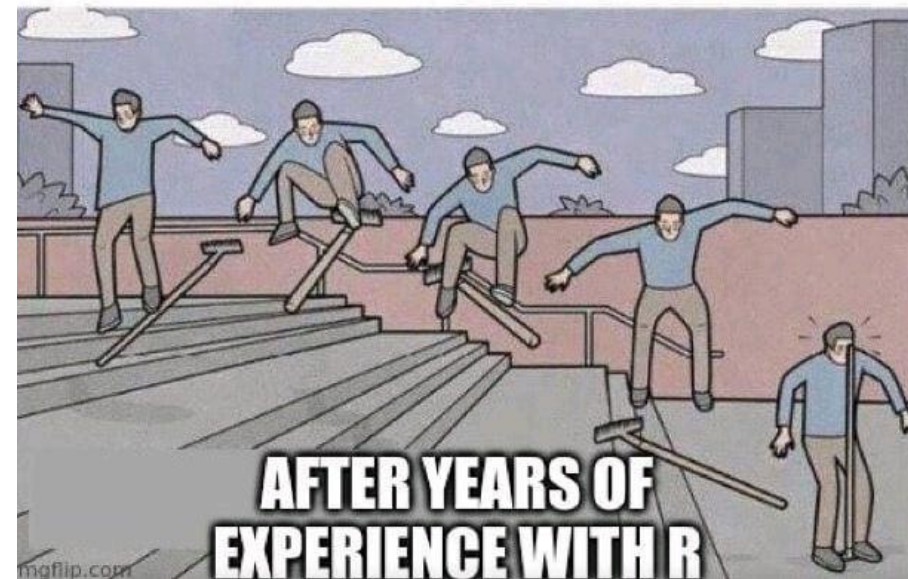
Stochastic character mapping

- Here complete character histories of the states are sampled from their probability distribution under a model
- States are predicted at both internal nodes AND along branches; computationally intensive



Now to some actual analyses

- Discrete trait mapping: tracking antigen turnover and geographic spread of *K. pneumoniae* SL1



Practical session

